

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO4-AT2 - Laboratory Performance | Programming in Java

Student Name	K. V. S. S. Jeswanth Babu
Register No.	192472012
Course Code	CSA0904
Course Name	Programming in Java
Assignment	CO4-AT2 - Laboratory Performance (Question 1)

QUESTION 1: SQUARE CALCULATOR APPLET (AWT CONTROLS)

Problem Statement:

Design an Applet that accepts a number from the user and displays its square when a button is clicked. Use AWT controls and appropriate event handling.

SOURCE CODE

```
import java.awt.*;
import java.awt.event.*;

// Main Class at the Top
public class SquareCalculatorApplet extends Frame implements ActionListener {

    private Label lblInput;
    private Label lblResult;
    private TextField txtInput;
    private Button btnCalculate;
    private Button btnClear;

    public SquareCalculatorApplet() {
        // AWT Window Setup & FlowLayout
        setTitle("Square Calculator (AWT)");
        setSize(360, 230);
        setBackground(new Color(240, 244, 248));
        setLayout(new FlowLayout(FlowLayout.CENTER, 15, 15));
        setLocationRelativeTo(null);

        // Header Label
        Label lblTitle = new Label("SQUARE CALCULATOR APPLET", Label.CENTER);
        lblTitle.setFont(new Font("Arial", Font.BOLD, 14));
        lblTitle.setForeground(new Color(30, 60, 114));
        add(lblTitle);

        // Input Field & Label
        lblInput = new Label("Enter a Number:");
        lblInput.setFont(new Font("Arial", Font.PLAIN, 12));
        add(lblInput);

        txtInput = new TextField(15);
        add(txtInput);
```

```

// AWT Buttons
btnCalculate = new Button("Calculate Square");
btnCalculate.setBackground(new Color(41, 128, 185));
btnCalculate.setForeground(Color.WHITE);
btnCalculate.addActionListener(this);
add(btnCalculate);

btnClear = new Button("Clear");
btnClear.setBackground(new Color(231, 76, 60));
btnClear.setForeground(Color.WHITE);
btnClear.addActionListener(this);
add(btnClear);

// Result Display Label
lblResult = new Label("Result: Waiting for input...", Label.CENTER);
lblResult.setFont(new Font("Arial", Font.BOLD, 12));
lblResult.setForeground(new Color(44, 62, 80));
add(lblResult);

// Close button handler
addWindowListener(new WindowAdapter() {
    @Override
    public void windowClosing(WindowEvent e) {
        dispose();
        System.exit(0);
    }
});

@Override
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == btnCalculate) {
        String inputStr = txtInput.getText().trim();

        if (inputStr.isEmpty()) {
            lblResult.setText("Error: Input cannot be empty!");
            lblResult.setForeground(Color.RED);
            return;
        }

        try {
            double number = Double.parseDouble(inputStr);
            double square = number * number;

            if (square == (long) square) {
                lblResult.setText("Square of " + (long) number + " = " + (long) square);
            } else {
                lblResult.setText("Square of " + number + " = " + String.format("%.4f", square));
            }
            lblResult.setForeground(new Color(39, 174, 96));

        } catch (NumberFormatException ex) {
            lblResult.setText("Error: Enter a valid numeric value!");
            lblResult.setForeground(Color.RED);
        }
    } else if (e.getSource() == btnClear) {
        txtInput.setText("");
        lblResult.setText("Result: Waiting for input...");
        lblResult.setForeground(new Color(44, 62, 80));
        txtInput.requestFocus();
    }
}

public static void main(String[] args) {
    new SquareCalculatorApplet().setVisible(true);
}

```

```
}  
}
```

OUTPUT SCREENSHOT

